

Python Practice Roadmap

Part 1 – Core Python (10 exercises)

1. **Even Numbers:** Given a list of integers, return a new list containing only the even numbers.
2. **Square Numbers:** Return the square of every number in a list (Solve using loop and list comprehension).
3. **Word Counter:** Count word frequencies in "python is great and python is easy".
4. **Largest Number:** Without using max().
5. **Second Largest Number:** Without sorting.
6. **Remove Duplicates:** From [1,2,2,3,4,4,5].
7. **Reverse Dictionary:** Swap keys and values.
8. **Merge Dictionaries:** Merge two dictionaries.
9. **Character Frequency:** Count every character.
10. **Palindrome:** Check if a string is a palindrome.

Part 2 – Functions (8 exercises)

1. Calculate average.
2. Return min and max together.
3. Accept any number of arguments (*args).
4. Function with optional parameters.
5. Return multiple values.
6. Lambda to sort students by GPA.
7. Use map() to convert list of strings to ints.
8. Use filter() to keep salaries above 8000.

Part 3 – Collections (8 exercises)

1. Flatten lists.
2. Common elements between two lists.
3. Difference between two sets.
4. Count unique words.
5. Sort dictionary by values.
6. Group students by department.
7. Find duplicates.
8. Rotate a list.

Part 4 – Files (6 exercises)

1. Read a text file.
2. Count words.
3. Count lines.
4. Find longest line.
5. Read CSV.
6. Write CSV.

Part 5 – Exceptions (4 exercises)

1. Handle division by zero.
2. Handle missing file.
3. Validate integer input.
4. Custom exception.

Part 6 – Python Tricks (8 exercises)

1. Use enumerate(), zip().
2. Sort list of tuples.
3. Dictionary/List/Nested comprehensions.

4. Generator expression.
5. Use with open().

Part 7 – Mini Data Tasks (6 exercises)

1. Average salary, Highest salary, Count per department, Duplicate IDs, Filter > 30.
2. Small report generation.

Final Mini Project: Employee Management System

Build a system to manage employee records (Add, Remove, Search, Statistics, Save/Load CSV).

Rules

- Write functions for every task.
- Use Git after every 5–10 exercises.
- If stuck, check documentation or ask for hints (avoid immediate full solutions).
- Focus on clean, readable code.